

PlanPO: Group Planning-Aware Policy Optimization for Multi-Turn Agentic LLMs

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Abstract

Group-relative policy optimization has emerged as a key paradigm for training agentic large language models (LLMs) on multi-turn interactive tasks. However, most existing variants fail to distinguish advantages among successful trajectories even when these trajectories differ substantially in their interaction efficiency. For instance, circuitous successes are often assigned the identical outcome reward, causing advantage collapse and severe performance bottlenecks. To this end, we propose *Group Planning-aware Policy Optimization (PlanPO)*, a simple yet effective RL method for learning generalizable planning abilities beyond task-specific high-quality behavior patterns. Specifically, PlanPO introduces coarse-to-fine advantage signals, which capture the relative differences in *trajectory-level* lengths and *turn-level* response lengths *conditioned on successful trajectories* sampled for the same task. Within the group-relative optimization structure, this enables agents to actively learn generalizable and deliberate behaviors spanning interaction planning and textual generation from high-quality rollouts, without degenerating into vanilla length minimization. Experimentally, PlanPO improves over GRPO by **27.2%** on average across the challenging multi-turn benchmarks ALFWorld, WebShop, and SciWorld, outperforming recent powerful baselines while incurring negligible additional training cost.

Introduction

Large Language Models (LLMs) have demonstrated remarkable progress across a wide range of complex multi-turn tasks, including information retrieval (Liao, Liao, and Gadiraju 2025; Eugene et al. 2026), web navigation (Rawles et al. 2025)(Zhang et al. 2026a), code generation (Zhang et al. 2024; Dai et al. 2026), and embodied interaction (Li et al. 2024; Zhang et al. 2026d). Recent work has increasingly explored agentic reinforcement learning (RL) with verifiable outcome rewards (Zhao et al. 2025a; Dai et al. 2026; Li et al. 2026), particularly Group Relative Policy Optimization (GRPO) (Shao et al. 2024), to fine-tune open-source LLMs such as Qwen2.5 (Yang et al. 2024), thereby improving the capabilities of LLM agents in multi-turn tasks.

However, a central challenge in on-policy agentic RL lies in *enriching the reward signals of rollout trajectories while improving data utilization* (Zhang et al. 2025a,b; Wang et al.

2026a; Peng et al. 2026). Many early studies addressed this issue by introducing value models, such as critics (Schulman et al. 2017; Dai et al. 2025) and process reward models (PRMs) (Chae et al. 2026), to evaluate turn-level behaviors. Yet these models may introduce estimation or proxy bias, and incur substantial memory costs (Chae et al. 2026). Recent group-optimization approaches instead mine informative rollouts to craft turn-level reward signals and achieve discriminative advantages. For example, HiPER (Peng et al. 2026) computes the multiple returns of trajectory segments via sub-task decomposition, while GiGPO (Feng et al. 2026) constructs step advantage signals over the action space by identifying repeated anchor states. Other methods, such as R3L (Shi et al. 2026) and GVPO (Zhang et al. 2026c), establish turn-level credit from failure reflection and diverse code execution feedback. However, most methods rely on labor-intensive manual design and empirical heuristics, which limits their generality across task settings. As a result, advantage collapse within successful rollout groups remains difficult to mitigate in a broadly applicable way (Zhao et al. 2026).

This motivates the question of how to enrich rollout-driven training signals for long-horizon tasks in a simple, effective, and more task-general manner. We begin by revisiting a naturally available yet underutilized signal in group rollouts, i.e., the length profiles across both turn-level interaction trajectories and token-level generated responses. Figure 1 provides an abstract illustration of this intuition. Specifically, our key observation is that many inefficiencies in agentic RL manifest as excessive interaction or generation length. In multi-turn interactions, agents may hesitate between states, repeatedly visit similar observations, or enter dead ends before eventually completing the task. A similar issue arises in token-level textual responses. Given the same question or turn-level observation, sampled responses may produce correct actions while still containing unnecessarily verbose, convoluted, or even logically flawed reasoning traces. Nevertheless, both inefficient turns and reasoning tokens can still share identical success rewards just like the superior solutions. Crucially, treating such heterogeneous successes as equally preferable weakens distinguishable signals and the underlying abilities, while allowing noisy rollouts to degrade training quality and impose substantial performance bottlenecks.

To address this, we present *Group Planning-aware Policy Optimization (PlanPO)*, an effective group-based RL method

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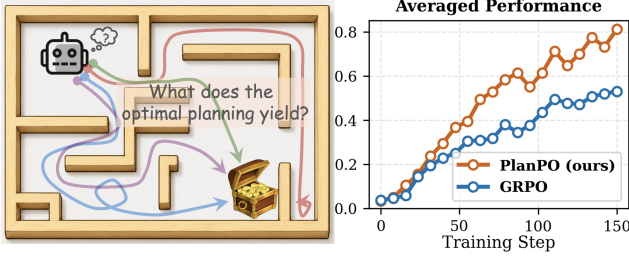


Figure 1: **Left:** Successful rollouts are not equally informative. Rollouts that reach the same task goal, can differ substantially in length, directness, and reasoning quality, while the most optimal strategies may reveal more sophisticated and generalizable capabilities. **Right:** Averaged normalization performance comparison across ALFWorld, WebShop, and SciWorld environments using the Qwen2.5-1.5b model.

for learning generalizable planning abilities beyond specific planful behaviors. Specifically, within the set of successful trajectories sampled in same task, PlanPO normalizes outcome rewards by turn-level trajectory lengths and token-level response lengths, constructing coarse-to-fine dense reward signals. Rather than directly summing these dense rewards, PlanPO then computes their relative advantages separately and combines them through a weighted formulation, preserving discriminative information at different granularities (Liu et al. 2026). Additionally, unlike generic length-based reward shaping (Pardo et al. 2018; Liu et al. 2025a), PlanPO performs multiscale length normalization only conditioned on successes with group-relative structure. Thus, successful completion is a prerequisite for group length normalization. Empirically, we show that these success-conditioned relative advantages help agents acquire planning behaviors that generalize beyond specific high-quality trajectories, rather than merely imitating task-specific success patterns.

We evaluate PlanPO on three challenging multi-turn benchmarks, ALFWorld (Shridhar et al. 2020), WebShop (Yao et al. 2022a), and SciWorld (Wang et al. 2022), using Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct. PlanPO consistently outperforms recent strong baselines while incurring negligible additional training cost, specifically improving over GRPO by **27.2%** on average and notably achieving a **24.3%** gain on out-of-distribution tasks in ALFWorld. Extensive ablations and analyses further validate its effectiveness and generalizability. Our contributions are threefold.

- We identify successful-rollout heterogeneity as a critical bottleneck in group-relative optimization, where redundant turns and reasoning can mask rollout quality and then limit policy learning.
- We propose PlanPO, which constructs success-conditioned length-normalized advantages, encouraging agents to learning planning-aware abilities beyond specific behavioral patterns.
- We show that PlanPO consistently outperforms recent strong baselines across ALFWorld, WebShop, and SciWorld, while incurring negligible additional training cost.

Preliminaries

Multi-Turn Agentic RL. We consider RL for LLM-based agents that interact with an environment over multiple turns, where the interaction process is formulated as a finite-horizon Markov Decision Process (MDP). Given a task instance $x \in p(X)$, at each turn $t = 1, 2, \dots, T$, an agent policy π_θ observes a state $s_t \in \mathcal{S}$ and generates a textual action $a_t \in \mathcal{A}$, and then transitions to the next state $s_{t+1} \in \mathcal{S}$ while yielding a scalar reward $r_t \in \mathbb{R}$. The interaction unfolds as a trajectory $\tau = \{(s_1, a_1, r_1), (s_2, a_2, r_2), \dots, (s_T, a_T, r_T)\}$, where T denotes the trajectory length in interaction turns. In practical tasks such as ALFWorld, the agent generates a response as its action for each observation. This response is structured within `<think></think>` and `<action></action>` tags, where the former contains the reasoning process, while the latter specifies the actual action executed in the environment. Notably, for most task settings, reward signals are sparse and delayed, e.g., the environment provides an outcome reward $R(\tau)$ only after the trajectory terminates.

Group-relative Policy Optimization

Recent agentic RL methods for LLMs commonly adopt a group-relative policy optimization paradigm. Given a task instance x , the old policy $\pi_{\theta_{\text{old}}}$ samples a group of N candidate trajectories $\mathcal{G}_x = \{\tau_1, \tau_2, \dots, \tau_N\}$, where each trajectory corresponds to one complete rollout. Each trajectory τ_i receives a scalar reward $R(\tau_i)$ that reflects the overall quality or success of the generated outcome. Instead of learning an advantage function $A(s_t, a_t)$ with critic networks in PPO (Schulman et al. 2017), group-based RL computes the advantage using only statistics within the sampled group:

$$A(\tau_i) = \text{GroupNormalization}(\{R(\tau_i)\}_{i=1}^N).$$

In GRPO (Shao et al. 2024), the advantage is evaluated by normalizing each trajectory reward with the mean and variance of group rewards ($\{R(\tau_i)\}_{i=1}^N$). This sampling-based estimator reduces the memory and computational overhead introduced by the critic architecture in conventional PPO.

Group Planning-Aware Policy Optimization for Multi-turn Agentic LLMs

We propose *group-relative Planning-aware Policy Optimization (PlanPO)*, a simple and effective group-relative RL method for learning high-level planning strategies beyond task-specific high-quality rollouts. We begin with our motivation, followed by the introduction of the coarse-to-fine advantage design, and conclude with the policy optimization loss and theoretical analysis.

Motivation

As illustrated in the right of Figure 2, given the same task, some rollouts reach the goal through short and coherent interaction paths, whereas others involve redundant state transitions or even enter loops and dead ends. At the turn level, verbose reasoning can also introduce inconsistencies or misleading intermediate claims. For example, as illustrated, given a task or turn observation, although the final sampled actions "`<action>take`"

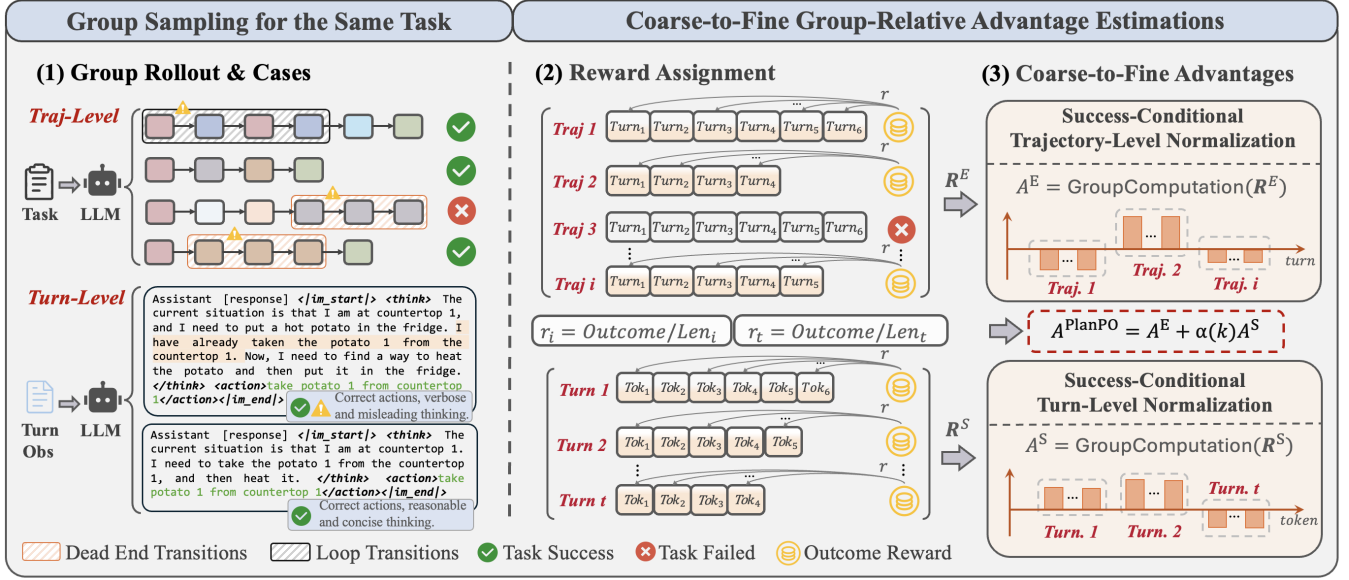


Figure 2: **Overview of PlanPO.** **Left:** Rollouts sampled for the same task can reach the same outcome through interaction paths and textual responses of markedly different quality. **Middle:** PlanPO converts the outcome into success-conditioned trajectory- and turn-level scores using trajectory and response lengths, respectively. **Right:** The two scores are normalized separately within the same-task group and combined into a coarse-to-fine advantage for policy optimization.

potato 1 from countertop 1</action>" are all correct, the thought process "I have already taken the potato ..." treats it as already happened. Such cases are common in the rollouts yet introduce hallucinated or logically inconsistent reasoning traces.

How should we quantify the quality of successful rollouts? Existing solutions may introduce post-hoc reflection (Shi et al. 2026) or agentic verifier models (Zhang et al. 2026b), but these signals are often costly and heuristic. The aforementioned observations motivate us to leverage multiscale rollout length as an initial signal. However, unconditional length-based reward shaping would degenerate into meaningless length minimization, even distorting the representation space of LLMs. Our goal is instead to comparatively learn planful behaviors within successful rollouts, while promoting task-general planning capabilities rather than fitting task-specific patterns. This objective naturally aligns with group-relative policy optimization. Below, we progressively construct the above conditional advantage signals.

Trajectory Length-Normalized Advantage

To achieve this, we first instantiate the success-conditioned length signal at the trajectory (or episode) level. For successful rollouts with the same outcome reward, trajectory length serves as a coarse proxy for planning efficiency across environment interactions. We therefore normalize the outcome reward by trajectory length only for successful rollouts, and compute the conditional group-relative advantage.

Formally, given a task instance x , the policy $\pi_{\theta_{\text{old}}}$ samples N trajectories $\{\tau_1, \tau_2, \dots, \tau_N\}$ start from the identical and initial state s_0 . Each rollout trajectory is represented as $\tau_i = \{(s_{i,1}, a_{i,1}, r_{i,1}), (s_{i,2}, a_{i,2}, r_{i,2}), \dots, (s_{i,T_i}, a_{i,T_i}, r_{i,T_i})\}$, where T_i represents the number of turns in i -th trajectory.

In our task setting, each trajectory receives only a terminal outcome reward $R(\tau_i) = 10$ when the task goal is reached. We then denote the trajectory-level group of sampled trajectories and rewards as:

$$\mathcal{G}_x^E = \{(\tau_1, R(\tau_1)), (\tau_2, R(\tau_2)), \dots, (\tau_N, R(\tau_N))\}, \quad (1)$$

where the superscript E denotes the episode, i.e., trajectory level. Let $\mathcal{G}_x^U \subseteq \mathcal{G}_x^E$ denote the subset of successful trajectories, and let $\mathbb{1}[\tau_i \in \mathcal{G}_x^U]$ indicate whether τ_i succeeds. For each trajectory, we define its length-normalized episode reward as,

$$R^E(\tau_i) = \mathbb{1}[\tau_i \in \mathcal{G}_x^U] R(\tau_i) / T_i, \quad (2)$$

where T_i is the number of valid interaction turns in τ_i . The corresponding trajectory-level advantage A^E is computed by group-relative normalization:

$$A^E(\tau_i) = \frac{R^E(\tau_i) - \text{mean}(\{R^E(\tau_j)\}_{j=1}^N)}{F_{\text{norm}}(\{R^E(\tau_j)\}_{j=1}^N)}. \quad (3)$$

Here, $F_{\text{norm}}(\cdot)$ is the normalization factor, instantiated as either $\text{std}(\cdot) + \epsilon$ or 1 (Feng et al. 2026), with the former usually used by default, where ϵ is a small constant for numerical stability (Shao et al. 2024). The trajectory-level $A^E(\tau_i)$ is broadcast to each turn in the i -th trajectory, providing a trajectory-wide discrimination signal among successful rollout trajectories that complete the same task but differ in process quality or long-term planning.

Response Length-Normalized Advantage

While the above advantage provides learning signals for discriminating trajectories, each response turn within a trajec-

tory still lacks fine-grained credit. Next, we compute group-relative advantages only among successful responses, so that the agent learns that planful responses are desirable only when the corresponding actions remain correct. Additionally, we flatten all successful responses generated for the same task into one group for advantage computation.

We then employ a similar group relative advantage structure to achieve the above idea. Formally, for the t -th interaction turn in trajectory τ_i , we collect all active responses into a turn-level group,

$$\mathcal{G}_{\mathbf{x},t}^S = \{(\mathbf{a}_{i,t}, R(\tau_i)) \mid \tau_i \in \mathcal{G}_{\mathbf{x}}^E, t \leq T_i\}, \quad (4)$$

where S denotes the step, i.e., turn level, and $L_{i,t} = |\mathbf{a}_{i,t}|$ is the token length of the response generated by trajectory τ_i at turn t . Similar to the trajectory-level case, we define the response length-normalized reward as,

$$R^S(\mathbf{a}_{i,t}) = \mathbb{1}[\tau_i \in \mathcal{G}_{\mathbf{x}}^U] R(\tau_i) / L_{i,t}. \quad (5)$$

The turn-level relative advantage is then computed within the active response group:

$$A^S(\mathbf{a}_{i,t}) = \frac{R^S(\mathbf{a}_{i,t}) - \text{mean}(\{R^S(\mathbf{a}_{j,t}) \mid \mathbf{a}_{j,t} \in \mathcal{G}_{\mathbf{x},t}^S\})}{F_{\text{norm}}(\{R^S(\mathbf{a}_{j,t}) \mid \mathbf{a}_{j,t} \in \mathcal{G}_{\mathbf{x},t}^S\})}. \quad (6)$$

Here, the resulting turn advantage $A^S(\mathbf{a}_{i,t})$ is assigned to all tokens in the response $\mathbf{a}_{i,t}$, which provides a fine-grained supervision signal for successful responses.

Coarse-to-Fine Group Policy Optimization

We integrate the trajectory-level and turn-level advantages into a multiscale group-relative advantage for policy optimization. For the response $\mathbf{a}_{i,t}$ in trajectory τ_i , the final PlanPO advantage is defined as,

$$A^{\text{PlanPO}}(\mathbf{a}_{i,t}) = A^E(\tau_i) + \alpha(k)A^S(\mathbf{a}_{i,t}), \quad (7)$$

$$\text{with } \alpha(k) = \text{LinearDecay}(k; \alpha_{\text{init}}, \alpha_{\text{final}}), \quad (8)$$

where $\alpha(k)$ denotes the decay weight with training step k for balancing the two level signals. We set the turn-level coefficient smaller than the trajectory-level coefficient, i.e., $0 \leq \alpha_{\text{final}} \leq \alpha(k) < \alpha_{\text{init}} \leq 1$. This design keeps trajectory-level planning quality as the dominant signal and uses response length signals only as a refinement. Although the response constraint is applied only to successful trajectories, an overly large $\alpha(k)$ experimentally over-penalize response length, thereby hurting task performance. Please see the ablation study for details. Therefore, we gradually decay $\alpha(k)$ during training to reduce the strength of response-length normalization as the policy becomes more capable.

Finally, PlanPO optimizes the policy with the same clipped group-relative objective backbone as GRPO-style methods:

$$\mathcal{J}^{\text{PlanPO}}(\theta) = \mathbb{E}_{\substack{\mathbf{x} \sim p(X), \\ \{\tau_i\}_{i=1}^N \sim \pi_{\theta_{\text{old}}}}} \left[\frac{1}{\sum_{i=1}^N T_i} \sum_{i=1}^N \sum_{t=1}^{T_i} \ell_{i,t} - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta}(\cdot \mid \mathbf{x}) \parallel \pi_{\text{ref}}(\cdot \mid \mathbf{x})) \right], \quad (9)$$

$$\ell_{i,t} = \min \left(\rho_{\theta}(\mathbf{a}_{i,t}) A^{\text{PlanPO}}(\mathbf{a}_{i,t}), \right. \\ \left. \text{clip}(\rho_{\theta}(\mathbf{a}_{i,t}), 1 - \epsilon, 1 + \epsilon) A^{\text{PlanPO}}(\mathbf{a}_{i,t}) \right). \quad (10)$$

Here, $\rho_{\theta}(\mathbf{a}_{i,t}) = \pi_{\theta}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}, \mathbf{x}) / \pi_{\theta_{\text{old}}}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}, \mathbf{x})$ is the importance sampling ratio, ϵ is the clipping coefficient, and β controls the strength of KL regularization with respect to the reference policy π_{ref} .

Theorem 1 (Bias–variance trade-off in PlanPO). *Consider a fixed response $\mathbf{a}_{i,t}$ at optimization step k , with the randomness induced by repeated same-task group sampling. Let $A^{E,*} = \mathbb{E}_{\mathcal{G}}[A^E]$ and $A^{S,*} = \mathbb{E}_{\mathcal{G}}[A^S]$ denote the expected trajectory-level signal and turn-level refinement, respectively. Assume that their centered estimation errors have variances v_E and v_S and non-negative covariance c . Relative to the full-refinement target $A^* = A^{E,*} + A^{S,*}$, the PlanPO estimator $A_{\alpha} = A^E + \alpha A^S$, where $\alpha \in [0, 1]$, satisfies*

$$\text{Bias}^2(A_{\alpha}) = (1 - \alpha)^2 (A^{S,*})^2, \\ \text{Var}(A_{\alpha}) = v_E + \alpha^2 v_S + 2\alpha c.$$

Consequently, increasing α decreases the squared bias while increasing the variance, with:

$$0 \leq \text{Bias}^2(A_{\alpha}) \leq (A^{S,*})^2, \\ v_E \leq \text{Var}(A_{\alpha}) \leq (\sqrt{v_E} + \sqrt{v_S})^2.$$

This theorem shows that $\alpha(k)$ relates to PlanPO’s bias–variance trade-off. A larger $\alpha(k)$ preserves more turn-level refinement but raises sampling variance, whereas a smaller $\alpha(k)$ reduces variance at the cost of greater shrinkage bias. The technical supplement provides detailed proof.

Experiments

We evaluate PlanPO across a range of multi-turn environments. Specifically, the experiments aim to answer three questions: (1) *How well does PlanPO perform overall across these challenging environments?* (2) *How does PlanPO achieve its performance gains?* (3) *Does PlanPO acquire the capabilities of task generalization and planning awareness rather than overfitting to task-specific success patterns?*

Experimental Setup

Benchmarks. We train and evaluate PlanPO on three challenging multi-turn benchmarks: ALFWorld (Shridhar et al. 2020), WebShop (Yao et al. 2022a), and SciWorld (Wang et al. 2022). ALFWorld is an embodied household environment for evaluating long-horizon textual reasoning and decision-making. In each episode, the agent receives a concrete task instruction sampled from 3,827 tasks from six categories. WebShop is an interactive web-shopping environment containing nearly 1.1 million products and 12,000 user instructions. SciWorld is designed for scientific tasks and provides APIs through which agents can manipulate scientific instruments and conduct experiments. The technical supplement provides further descriptions of the benchmarks.

Type	Method	ALFWorld							WebShop	
		Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
Closed-Source Model										
Prompting	GPT-4o	75.3	60.8	31.2	56.7	21.6	49.8	48.0	31.8	23.7
Prompting	Gemini-2.5-Pro	92.8	63.3	62.1	69.0	26.6	58.7	60.3	42.5	35.9
Qwen2.5-1.5B-Instruct										
Prompting	ReAct	17.4	20.5	15.7	6.2	7.7	2.0	12.8	40.1	11.3
Prompting	Reflexion	35.3	22.2	21.7	13.6	19.4	3.7	21.8	55.8	21.9
RL Training	PPO (with critic)	64.8 $\pm$ 3.5	40.5 $\pm$ 6.9	57.1 $\pm$ 4.9	60.6 $\pm$ 6.6	46.4 $\pm$ 4.0	47.4 $\pm$ 1.9	54.4 $\pm$ 3.1	73.8 $\pm$ 3.0	51.5 $\pm$ 2.9
RL Training	RLOO	88.3 $\pm$ 3.0	52.8 $\pm$ 8.6	71.0 $\pm$ 5.9	62.8 $\pm$ 8.7	66.4 $\pm$ 5.5	56.9 $\pm$ 4.7	69.7 $\pm$ 2.5	73.9 $\pm$ 5.6	52.1 $\pm$ 6.7
RL Training	GRPO	85.3 $\pm$ 1.5	53.7 $\pm$ 8.0	84.5 $\pm$ 6.8	78.2 $\pm$ 7.9	59.7 $\pm$ 5.0	53.5 $\pm$ 5.6	72.8 $\pm$ 3.6	75.8 $\pm$ 3.5	56.8 $\pm$ 3.8
RL Training	EMPG	85.5	33.5	78.9	76.2	74.7	89.1	73.7	80.4	60.8
RL Training	GiGPO _{w/ std}	94.4 $\pm$ 5.9	67.5 $\pm$ 4.6	94.8 $\pm$ 3.8	94.4 $\pm$ 7.8	79.8 $\pm$ 4.7	76.4 $\pm$ 5.4	86.7 $\pm$ 1.7	83.1 $\pm$ 1.6	65.0 $\pm$ 3.2
RL Training	GiGPO _{w/o std}	96.0 $\pm$ 1.4	76.5 $\pm$ 3.9	91.8 $\pm$ 5.5	91.3 $\pm$ 6.3	71.7 $\pm$ 8.4	79.5 $\pm$ 7.7	86.1 $\pm$ 4.7	83.5 $\pm$ 1.8	67.4 $\pm$ 4.5
RL Training	PlanPO	98.2 $\pm$ 1.1	85.1 $\pm$ 4.6	94.6 $\pm$ 4.7	93.8 $\pm$ 6.0	82.4 $\pm$ 6.7	83.7 $\pm$ 8.0	91.3 $\pm$ 4.1	86.8 $\pm$ 1.5	77.2 $\pm$ 5.6
Qwen2.5-7B-Instruct										
Prompting	ReAct	48.5	35.4	34.3	13.2	18.2	17.6	31.2	46.2	19.5
Prompting	Reflexion	62.0	41.6	44.9	30.9	36.3	23.8	42.7	58.1	28.8
RL Training	PPO (with critic)	92.3 $\pm$ 4.0	64.0 $\pm$ 8.4	92.5 $\pm$ 2.4	89.5 $\pm$ 7.0	80.3 $\pm$ 2.0	68.8 $\pm$ 8.3	80.4 $\pm$ 2.7	81.4 $\pm$ 3.1	68.7 $\pm$ 5.1
RL Training	RLOO	87.6 $\pm$ 4.3	78.2 $\pm$ 8.3	87.3 $\pm$ 5.8	81.3 $\pm$ 7.6	71.9 $\pm$ 5.2	48.9 $\pm$ 8.4	75.5 $\pm$ 4.6	80.3 $\pm$ 3.2	65.7 $\pm$ 4.0
RL Training	GRPO	90.8 $\pm$ 5.1	66.1 $\pm$ 6.7	89.3 $\pm$ 5.4	74.7 $\pm$ 6.9	72.5 $\pm$ 5.4	64.7 $\pm$ 7.3	77.6 $\pm$ 5.2	79.3 $\pm$ 2.8	66.1 $\pm$ 3.7
RL Training	EMPG	92.9	75.2	74.8	86.3	73.7	65.3	78.5	81.0	69.3
RL Training	GiGPO _{w/ std}	97.7 $\pm$ 1.6	82.7 $\pm$ 7.9	98.8 $\pm$ 1.6	83.7 $\pm$ 7.2	89.3 $\pm$ 8.2	79.2 $\pm$ 6.6	90.8 $\pm$ 1.3	84.4 $\pm$ 2.9	72.8 $\pm$ 3.2
RL Training	GiGPO _{w/o std}	91.8 $\pm$ 5.4	88.6 $\pm$ 6.3	95.9 $\pm$ 3.2	90.2 $\pm$ 2.6	86.5 $\pm$ 5.5	85.2 $\pm$ 7.5	90.2 $\pm$ 2.3	86.2 $\pm$ 2.6	75.2 $\pm$ 3.8
RL Training	PlanPO	100.0 $\pm$ 0.0	92.1 $\pm$ 6.3	96.8 $\pm$ 2.5	99.0 $\pm$ 1.4	88.3 $\pm$ 6.8	89.7 $\pm$ 7.8	94.4 $\pm$ 2.1	88.5 $\pm$ 3.2	80.6 $\pm$ 4.7

Table 1: Evaluation results on ALFWorld and WebShop. For each RL training method, we report the mean and variance over three random seeds. The ALFWorld contains six categories: Pick & Place (Pick), Examine in Light (Look), Clean & Place (Clean), Heat & Place (Heat), Cool & Place (Cool), and Pick Two & Place (Pick2). Most entries in this table are reported by Feng et al. (Feng et al. 2026). Notably, the baseline GiGPO_{w/ std} denotes $F_{\text{norm}} = \text{std}$, and GiGPO_{w/o std} denotes $F_{\text{norm}} = 1$.

Baselines. For ALFWorld and WebShop, we compare PlanPO against three categories of competitive baselines: (1) Proprietary models: GPT-4o (Achiam et al. 2023) and Gemini-2.5-Pro (Team et al. 2023); (2) training-free prompting agents: ReAct (Yao et al. 2022b) and Reflexion (Shinn et al. 2023); and (3) RL-based methods: PPO (Schulman et al. 2017), RLOO (Ahmadian et al. 2024), GRPO (Shao et al. 2024), EMPG (Wang et al. 2026b), and GiGPO (Feng et al. 2026), which cover representative actor-critic and group-relative optimization approaches. For SciWorld, we consider (1) training-free agents based on the OpenAI (Achiam et al. 2023), Gemini (Team et al. 2023), Qwen (Yang et al. 2025), Llama (Grattafiori et al. 2024), and DeepSeek (Liu et al. 2024; Guo et al. 2025) model families, spanning both proprietary models and open-source models of different scales; and (2) RL-based methods: AgentGym-RL (Xi et al. 2026), a GRPO framework for training across multiple environments, and ScalingInter (Xi et al. 2026), a group-based RL method that reports strong performance on SciWorld.

Implementation details. We employ Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct as the base models for training experiments. To ensure fair comparisons, the hyperparameter settings of PlanPO follow the existing RL framework (Feng et al. 2026; Xi et al. 2026) in each benchmark. Specifically, we use a group size of $N = 8$, a learning rate of 1×10^{-6} , and a KL-penalty coefficient of 1×10^{-3} for Sci-

World and 0.01 for the other environments. The maximum number of interaction turns is set to 50 for ALFWorld, 15 for WebShop, and 20 for SciWorld. We train for 150 steps on ALFWorld and WebShop and for 200 steps on SciWorld. On ALFWorld and WebShop, we set the turn-level weight $\alpha_{\text{init}} = 2\alpha_{\text{final}} = 0.1$ in PlanPO. For the exploratory SciWorld, we reduce it to 0.05 without further tuning to improve training stability. Finally, all our experiments were run on 6 NVIDIA H200s and 8 NVIDIA A40s. The technical supplement provides additional implementation details.

Experiment Results

Performance on ALFWorld and WebShop. As shown in Table 1, PlanPO substantially outperforms advanced closed-source models on both benchmarks. For the open-source Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct models, RL methods are significantly stronger than prompting-based methods. Nevertheless, PlanPO also outperforms existing powerful baselines, e.g., GiGPO and EMPG. For example, on the 1.5B model, PlanPO achieves an overall success rate of 91.3% across the six ALFWorld categories, improving over GRPO by +18.5 points, while requiring only several regularization operations with negligible computational cost.

Performance on SciWorld. In Table 2, PlanPO achieves consistently the best overall scores on complex and exploratory scientific scenarios, notably improving the GRPO-

Type	Model	Measure	Test-Cond.	Find	Chem-Mix	Lifespan	Overall
Proprietary	GPT-4o	15.09	6.02	38.64	20.00	73.33	21.00
	Qwen-Max	9.43	0.00	34.09	20.00	40.00	13.50
	Gemini-2.5-Flash	11.32	0.00	54.55	0.00	80.00	21.00
	OpenAI o4-mini	20.75	14.46	47.73	0.00	100.00	29.50
	OpenAI o3	47.17	25.30	56.82	40.00	66.67	41.50
	Gemini-2.5-Pro	9.43	0.00	29.55	0.00	46.67	12.50
Models ≥ 100 B	Qwen3-235B-A22B	11.32	4.82	59.09	20.00	66.67	23.50
	DeepSeek-V3-0324	0.00	0.00	2.27	0.00	0.00	0.50
	DeepSeek-R1-0528	1.89	0.00	11.36	0.00	20.00	4.50
Models < 100 B	Qwen2.5-72B-Instruct	7.55	1.20	15.91	20.00	40.00	9.50
	Qwen3-32B	5.66	1.20	31.82	0.00	66.67	14.00
	Llama-3.1-70B-Instruct	24.53	4.82	40.91	40.00	86.67	25.00
RL Training	AgentGym-RL-3B	20.75	28.92	0.00	0.00	66.67	22.50
	AgentGym-RL-7B	24.53	59.04	65.91	0.00	66.67	50.50
	ScalingInter-7B	33.96	55.42	88.64	0.00	73.33	57.00
	PlanPO-7B	40.52	69.38	91.67	0.00	85.94	68.46

Table 2: Evaluation results on SciWorld over three random seeds. The task Test-Cond. means test conductivity and Chem-Mix means chemistry mix. Baseline entries are reported by Zhang et al. (Zhang et al. 2026b).

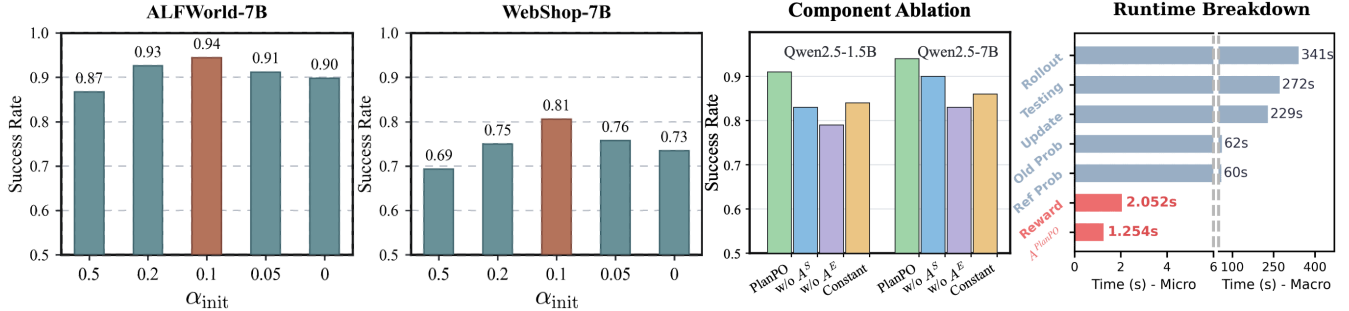


Figure 3: **Left two panels:** Ablation study on the weight schedule $\alpha(k)$ with $\alpha_{init} = 2\alpha_{final}$ in PlanPO. **Third panel:** Component ablation. We compared the ablation results of PlanPO with default decay, trajectory-level advantage A^E only, turn-level advantage A^S only, and PlanPO with constant $\alpha = 0.1$. **Right panel:** Runtime breakdown of each PlanPO iteration on four NVIDIA A40 GPUs. Compared to the GRPO, the additional computation introduced by PlanPO lies in the reward and advantage stages, which is negligible. Conversely, PlanPO can reduce overall task runtime by 12.5%

based AgentGym-RL from 50.5 to 68.46. However, we observe that RL training methods exhibit a consistent failure pattern in the Chem-Mix domain, which may be due to the model’s limited understanding and exploration ability required for rigorous scientific analysis (Xi et al. 2026).

In-depth Analysis

Ablation Study. To verify how PlanPO achieves performance gains, we performed two ablation studies on the scheduling coefficient $\alpha(k)$ and each component. As shown in left Figure 3, we compared a series of initial $\alpha_{init} = \{0.5, 0.2, 0.1, 0.05, 0\}$. We observed that $\alpha_{init} = 0.1$ achieved the best performance, while both smaller values (e.g., $\alpha_{init} = 0$) and excessively large values (e.g., 0.5) degraded performance. This matches the theoretical bias-variance trade-off, i.e., the turn-level signal is useful as a refinement, but should not dominate the trajectory-level plan-

ning signal. In the third panel of Figure 3, we observe that ablating any level signal significantly degrades PlanPO’s performance, especially without A^E . Regarding turn-level advantages, PlanPO with coefficient decay settings consistently outperforms those using constant coefficients. In summary, its performance gains primarily stem from the A^E , while a properly weighted A^S further enables optimal performance.

Generalization Verification. The gains of PlanPO remain strong when the evaluation tasks differ from the training distribution. As shown in Table 3, PlanPO achieves 87.1% success on out-of-distribution ALFWorld tasks, improving over GRPO by +17.0 points and over GiGPO by +4.7 points. The gap between in-distribution and out-of-distribution performance is also modest, decreasing from 91.3% to 87.1%. This suggests that group-based PlanPO does not merely memorize some better planning templates, but learns planning-aware

Type	Method	In-Success	Out-Success
Prompting	GPT-4o	48.0	46.0
Prompting	Gemini2.5	60.3	50.5
RL Training	PPO	54.4 $\pm$ 3.1	50.9 $\pm$ 7.6
RL Training	RLOO	69.7 $\pm$ 2.5	68.7 $\pm$ 10.7
RL Training	GRPO	72.8 $\pm$ 3.6	70.1 $\pm$ 2.5
RL Training	GiGPO	86.7$\pm$1.7	82.4$\pm$2.0
RL Training	PlanPO	91.3$\pm$4.1	87.1$\pm$3.6

Table 3: Generalization evaluation with three seeds in ALFWorld using Qwen2.5-1.5B-Instruct. Some entries are reported by He et al. (He et al. 2026). The In-Success reports the success rate on the in-distribution tasks, while the Out-Success reports the success rate on out-of-distribution tasks.

abilities that generalize to unseen task configurations.

Length-Normalized Method Comparison. Figure 4 further isolates the role of success-conditioned length normalization. Directly applying unconditional length shaping in PPO quickly collapses to nearly zero success, showing that shorter generations or trajectories are not intrinsically better. Success conditioning alone is also insufficient, as PPO with success-conditioned length shaping remains close to standard PPO. By contrast, PlanPO steadily improves after training begins and reaches around 0.55 success near the end, clearly above GRPO and PPO variants. These results indicate that the key benefit comes from coupling length-induced signals with group-relative policy optimization, which turns heterogeneous successful rollouts into useful supervision. Note that the reward shaping operation and outcome reward size $R(\tau) = 10$ involved are consistent with PlanPO settings, except for the condition control and basic algorithm.

Planning-Aware Strategies. We further examine whether PlanPO induces planning-aware behavior beyond improving final task success. As shown in Figure 5, PlanPO consistently produces shorter trajectories than GRPO during training, reducing the mean number of interaction turns to 13.8 compared with 26.1 for GRPO. This suggests that PlanPO learns to reach task goals through more direct interaction paths, avoiding redundant transitions, repeated trials, and unnecessary detours. At the response level, PlanPO also yields more compact generations, with the mean response length decreasing to 56.3 tokens, while GRPO remains substantially more verbose at 95.1 tokens. Importantly, this behavior does not arise from naive length minimization: as shown in Figure 4, unconditional length shaping hurts task success, whereas PlanPO applies length normalization only under success-conditioned group-relative comparisons. These results indicate that PlanPO encourages planning-aware strategies by preferring successful rollouts that are both interaction-efficient and response-concise, thereby turning heterogeneity among successful trajectories into a useful signal for learning generalizable agentic planning abilities.

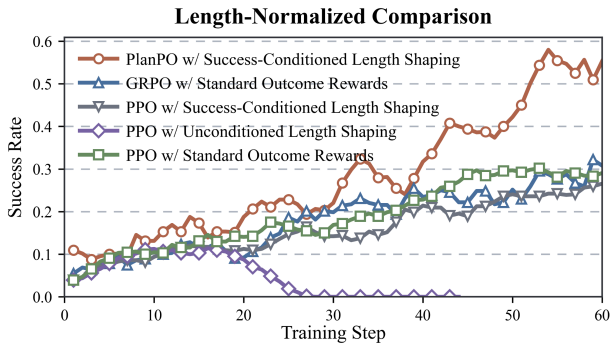


Figure 4: Length-normalized Comparison Analysis with different reward shaping settings in ALFWorld.

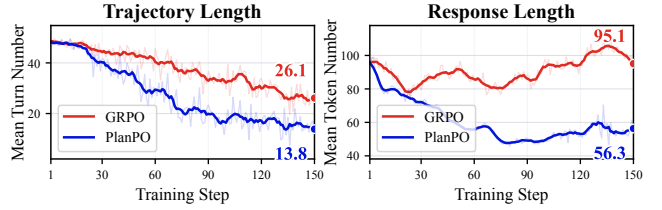


Figure 5: Mean Length Comparison in ALFWorld tasks.

ing to 56.3 tokens, while GRPO remains substantially more verbose at 95.1 tokens. Importantly, this behavior does not arise from naive length minimization: as shown in Figure 4, unconditional length shaping hurts task success, whereas PlanPO applies length normalization only under success-conditioned group-relative comparisons. These results indicate that PlanPO encourages planning-aware strategies by preferring successful rollouts that are both interaction-efficient and response-concise, thereby turning heterogeneity among successful trajectories into a useful signal for learning generalizable agentic planning abilities.

Related Work

RL has become a central recipe for improving LLM reasoning and interactive agents (Sheng et al. 2025; Shao et al. 2024; Zeng et al. 2026; Team et al. 2026). PPO-based RLHF (Schulman et al. 2017) trains a value model to reduce variance, while lighter critic-free alternatives such as ReMax (Li et al. 2023), RLOO (Ahmadian et al. 2024), and GRPO (Shao et al. 2024) replace the learned critic with sampled baselines or group-relative rewards, improving scalability for LLMs. Recent GRPO variants further refine this objective (Liu et al. 2025a; Cui et al. 2025; Zheng et al. 2025; Lin et al. 2026). DAPO improves stability with decoupled clipping, dynamic sampling, and overlong-response shaping (Yu et al. 2026). Dr. GRPO analyzes length and reward-variance normalization biases (Liu et al. 2025b), and GMPO stabilizes updates by changing the aggregation of token-level rewards (Zhao et al. 2025b). In parallel, agentic RL methods extend outcome-reward optimization to long-horizon interaction, including WebSailor (Li et al. 2025), GiGPO (Feng et al. 2026), HGPO (He et al. 2026), and A2TGPO (Chen et al. 2026), which study credit assignment through episode-, state-, history-, or turn-grouped advantages. Different from these methods, our work focuses on planning-aware credit assignment under successful-rollout heterogeneity.

Conclusion

This paper addresses successful-rollout heterogeneity in group-relative RL for multi-turn agentic LLMs. We propose PlanPO, which constructs success-conditioned length-induced advantages from trajectory-level interaction and token-level response generation. The experiments across multiple benchmarks demonstrate that PlanPO consistently outperforms existing methods and achieves the best task performance with stronger generalization, planful strategies, and negligible overhead. Our results suggest that successful-rollout heterogeneity provides a simple and scalable signal for learning planning-aware and generalizable behaviors.

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